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Secure Satellite Communication: Performance and cryptographic analysis of chaotic injected AES resistant to side-channel attacks targeting the S-box via machine unlearning.

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Contents

1 Introduction **3**

2 Background and State-of-the-Art of Satellite Communications **4**

2.1 SATCOM Architecture 5

2.1.1 Space Segment 6

2.1.2 Ground Segment 7

2.1.3 Link Segment 7

2.2 SATCOM Threats 8

2.2.1 Space Segment Threats 8

Bibliography **9**

Chapter 1

Introduction

Chapter 2

Background and State-of-the-Art of Satellite Communications

Satellite communications (SATCOM) are vastly different from how they appeared just a few decades ago. At that time, space-based communications were mainly represented by government activities, characterized by larger and more expensive satellites than the ones used today. This era is also known as the *old space* [1, pp. 377], a term often used to describe the space applications that emerged during the Cold War.

On the other hand, since the beginning of the *new space* era [1, pp. 377–398], thanks to the ever-decreasing cost of satellite launches and the enormous technological progress that has been made in recent years, space has become increasingly populated by satellites used for commercial purposes, such as:

- *Telecommunications and broadband connectivity*: they help to provide an internet connection even in the most remote regions or those that are hard to reach with a physical connection [1, pp. 373–381].
- *Television broadcast and multimedia streaming*: due to the constant demand for high-quality multimedia, the wide coverage of satellite communications helps to deliver content to a large number of users more efficiently [1, pp. 373–381].
- *Global Positioning System (GPS)*: numerous sectors make use of satellite-based geolocation to plan routes or monitor the movement of resources. In addition to this, a large number of people use geolocation daily to plan their travels [1, pp. 378–381].
- *Earth observation*: thanks to more affordable satellites, it is now possible for private entities to start their own Earth observation programs for scientific purposes, such as environmental analysis. Moreover, satellites observing the Earth are widely used for meteorological purposes [1, pp. 372–381].
- *Internet of Things (IoT)*: because of the multifaceted nature of IoT devices, which can take on many different forms, from smart homes to means of transport, it is essential that the coverage area serving these devices is as wide as possible. Such needs are effectively addressed by using SATCOMs as the main communication method [1, pp. 372].

- *Satellite as a service (SataaS)*: while it is true that the cost of satellite lounches is significantly decreased, it remains unaffordable for many private companies that would otherwise benefit of the wide coverage provided by SATCOMs.
  To address this challange, majour caerospace operators can lease resources on their satellites to third-party users, granting them access to space infrastructure without the need for additional launches [2].

Along with the opportunities offered by the *new space*, the critical requirements of the *old space* have not diminished. Today, satellite communications remain essential for handling mission-critical data exchanges, such as those required by aviation and maritime operations [1, pp. 397–398], as well as for the coordination of rescue services in disaster or crisis situations [1, p. 388–389].

Given the critical nature of the communications handled by SATCOM systems, it is becoming increasingly important to protect the information shared within these networks from malicious actors. To fully understand the scale of these security challenges and how threats can compromise the network, it is essential to analyze its underlying infrastructure. A modern SATCOM system is not a monolithic entity, but rather a complex, distributed architecture. Breaking down this infrastructure into its fundamental segments is a crucial step to accurately identify the specific attack surfaces that adversaries might exploit.

2.1 SATCOM Architecture

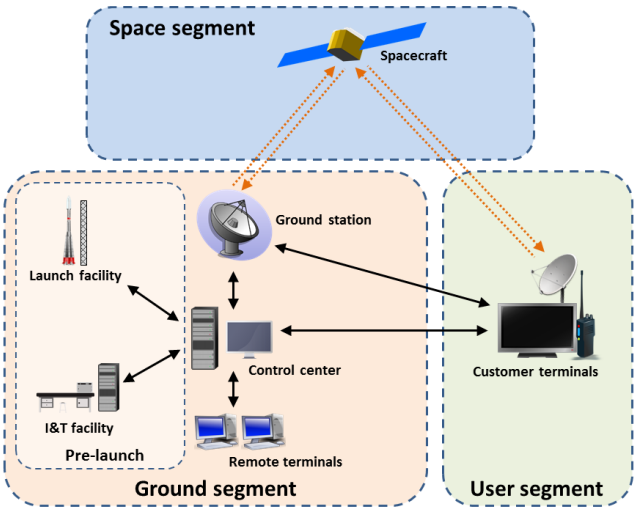


Figure 2.1:
Architectural overview of space, ground, and user segments in satellite systems [3].

Satellite communications are typically classified into three fundamental segments, although the exact categorization may vary across the literature. One common classification framework, as described in [1, p. 381], divides the architecture into the *Space Segment*, the

Ground Segment, and the *Link Segment*. In this model, the *Ground Segment* encompasses both the *Ground Stations*, which act as gateways for devices lacking direct satellite communication capabilities, and the *User Segment*, which consists of end-user terminals equipped for space links. Another classification model still includes the *Space Segment* and the *Ground Segment*, but explicitly separates the *User Segment*, treating the *Link Segment* merely as a passive communication channel [4, p. 5671].

While all these components are crucial to maintain the network's operability, this thesis will adopt the classification proposed in [1]. Grouping the ground stations and end-user devices under a unified *Ground Segment* is conceptually sound for terrestrial entities. More importantly, treating the *Link Segment* as a primary, distinct architectural component is essential for the scope of this study, as the physical transmission channel introduces specific vulnerabilities and security challenges that are central to the analyzed threat model.

2.1.1 Space Segment

The space segment is represented by the satellites in orbit, each composed of the *bus*, which handles onboard systems such as power and thermal management, and the *payload*, which is the part of the spacecraft designed to fulfill the satellite's specific mission [1, pp. 381–383]. Examples of payloads include telescopes to analyze distant objects or sensors to measure spatial properties.

Depending on the satellite's mission, its orbital height from Earth can vary radically [4]:

- *Low Earth Orbit (LEO)*: due to their low altitude, usually between 500 and 1200 km, these types of satellites are ideal for extending terrestrial networks thanks to their low latency (around 30 ms). However, because they are so close to the Earth, it is often necessary to deploy a group of interconnected *LEO* satellites, known as a *satellite constellation*, in order to cover a wider area of the planet. Additionally, because of the high risk of collision with space debris, the lifetime of *LEO* spacecraft is around 5 to 7 years, which is usually shorter than that of other satellites.

- *Medium Earth Orbit (MEO)*: usually positioned at an altitude of 2000 to 20000 km, they represent a midpoint between coverage and latency (around 125 ms). A significant share of *MEO* spacecraft function as GPS satellites, and typically 6 units are necessary in order to provide adequate service. Because of their intermediate position, *MEO* satellites often serve as relay nodes between *LEO* and *GEO* satellites. Finally, their operational lifetime is estimated at 10 to 15 years, depending on their altitude and exposure to radiation.

- *Geostationary Equatorial Orbit (GEO)*: operating in the highest orbital layer at approximately 36000 km, these spacecraft take exactly 24 hours to complete an orbit, appearing stationary to an observer on Earth. Thanks to their high altitude, they can

cover one-third of the entire planet's surface, making them ideal for broadcasting. However, it is important to account for the latency accumulated during transmission (around 600 ms). Due to their distance from the planet and the large quantity of resources needed to launch *GEO* spacecraft, they are usually built to last longer, typically around 15 to 20 years.

2.1.2 Ground Segment

This segment represents the terrestrial infrastructure that communicates with satellites. Depending on the role of the receiver or sender, it can be classified into several sub-components, the primary ones being [1, pp. 383–384]:

- *Ground Stations (GS)*: equipped with dedicated hardware, they can transmit and receive signals even from the most distant satellites. A vast amount of traffic passes through them, as the majority of terrestrial devices lack the necessary peripherals to communicate directly with spacecraft.
- *Mission Control Centers (MCC)*: their role is to ensure the satellite's mission continues undisturbed by analyzing the telemetry sent by the spacecraft. Telemetry consists of messages transmitted by the satellites containing critical information, such as their exact orbit or internal hardware measurements; this data allows operators to detect anomalies and perform necessary corrective actions.
- *User Segment*: this component comprises all the end-user devices, which receive messages after they are forwarded by the *Ground Stations*. Moreover, devices equipped with the necessary peripherals to communicate directly with the *Space Segment*, such as ships, airplanes, and Satellite TV, also fall into this category.

2.1.3 Link Segment

This segment is represented by the radio waves that link the Earth to space and the satellites to one another, as well as the hardware components used to establish these connections. Specifically, the *Link Segment* can be categorized as follows [1, pp. 381]:

- *Ground-to-Ground*: connects the *Ground Stations* to the *User Segment* and vice versa.
- *Ground-to-Space*: also known as the *uplink*, it represents the communication path from the *Ground Segment* to the *Space Segment*.
- *Space-to-Ground*: also known as the *downlink*, it represents the communication path from the *Space Segment* to the *Ground Segment*.
- *Space-to-Space*: connects the spacecraft within the *Space Segment* to one another, even across different orbital altitudes.

2.2 SATCOM Threats

Due to the distributed architecture of SATCOM networks, security threats targeting any individual segment must be rigorously addressed. Tampering with a single component can have cascading effects, jeopardizing the other segments and compromising both the spacecraft's mission and the broader network to which it is connected. Some threats associated with each segment are outlined below.

2.2.1 Space Segment Threats

While spacecraft are physically out of reach for potential attackers, this physical isolation does not exempt the *Space Segment* from cyber threats. Examples include [1, pp. 386–389]:

- *Software Attacks*: malicious software could be introduced into the satellite's systems via the *Link Segment*. Alternatively, in a *Satellite-as-a-Service (SataaS)* architecture, an attacker could exploit the shared infrastructure by gaining access to the spacecraft's resources under the guise of a legitimate user.
- *Denial of Service (DoS)*: these attacks pose a significant threat to the *Space Segment*, as an excessive volume of requests sent to the targeted satellite could overwhelm its processing capabilities, temporarily or permanently disrupting its mission.
- *Hardware attacks*: supply chain

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